ult2s4.py: Porting Ultimate Animations to Smash 4

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Overview

ult2s4.py turns Smash Ultimate animations into Smash 4 (Wii U) animations and writes them straight into the fighter's motion .pac files. It is the reverse of JoshuaSawagi's S4-Anims-to-Ult, in one Python file.

- **Input:** Ultimate animations exported from StudioSB as Maya `.anim` files, plus the Smash 4 fighter's `model.vbn` and its `motion/body/*.pac` files.
- **Output:** copies of every .pac with the chosen motions replaced, the loose `.omo` files, and a `report.txt` listing what changed.
- **Untouched:** every other animation in the .pac files stays byte-for-byte identical to the input.

Tested on Daisy's Ultimate animations onto Smash 4 Peach. Daisy's dash comes out within 0–5° of Peach's stock dash on every body bone.

Requirements

You need Python 3 with numpy, StudioSB, and the files below from both games.

What	Where it comes from	Notes
Python 3	python.org	Any recent version works
numpy	`pip inscould put this into tall numpy`	The only extra library
StudioSB	Smash Ultimate modding tool	Opens Ultimate models and exports animations as `.anim`
Ultimate fighter files	Extracted from Ultimate (`fighter/<name>/model/body/c00` and `fighter/<name>/motion/body/c00`)	The model so StudioSB has a skeleton, and the `.nuanmb` animations

What	Where it comes from	Notes
Smash 4 `model.vbn`	`fighter/<name>/model/body/c00/model.vbn` (any slot with the stock skeleton)	Gives the bone IDs and rest positions
Smash 4 motion .pac files	`fighter/<name>/motion/body/` from your Sm4shExplorer extract or workspace	main.pac, lhand.pac, rhand.pac and the swing pacs (e.g. peachsw1.pac)

Step 1: Export the Ultimate animations

Each Ultimate animation you want becomes one `.anim` file, exported from StudioSB with every bone included.

1. Open the Ultimate fighter's model in StudioSB (the `model/body/c00` folder), so the skeleton is loaded.
2. Load the `.nuanmb` animation from `motion/body/c00` (for example `a00wait1.nuanmb`).
3. Export it as a Maya `.anim` file. Include all bones and keep every frame.
4. Name the file after the motion, like `a00wait1.anim` or `a02run.anim`. The script reads the Smash 4 motion name from the file name.
5. Repeat for each animation, and put all the `.anim` files in one folder.

Rotations in the `.anim` can be in degrees or radians; the script reads the file's own unit setting and converts.

Step 2: Gather the Smash 4 files

The script needs the fighter's skeleton file and one folder holding its motion .pac files.

- **Skeleton:** copy `model.vbn` from `fighter/<name>/model/body/c00`. A custom costume's VBN works too, as long as it uses the stock skeleton.
- **Motion pacs:** copy everything in `fighter/<name>/motion/body/` into its own folder, such as `pacs/`. For Peach that is `main.pac`, `lhand.pac`, `rhand.pac`, `peachsw1.pac` and `peachsw2.pac`.
- **Starting point:** use the stock pacs from your extract for a clean start. Use your workspace pacs instead if they already hold animations you want to keep; only the motions you name get replaced.

- **Folder contents:** keep only that fighter's .pac files in the folder. Every .pac in it is read and written back out.

Sm4shExplorer's workspace folders are nested deep, so copying these files to a short path (like a folder in Downloads) makes the command easier to type.

Step 3: Run the script

One command converts any number of animations at once. Open a Command Prompt in the folder that holds `ult2s4.py` and run:

```
python ult2s4.py --vbn "C:\port\model.vbn" --pacs "C:\port\pacs" --out
"C:\port\out" "C:\port\anims\a00wait1.anim" "C:\port\anims\a02run.anim"
```

Argument	Required	What it does
`--vbn`	Yes	The Smash 4 `model.vbn` from Step 2
`--pacs`	Yes	The folder holding the fighter's motion .pac files
`--out`	Yes	Where the new .pac files, .omo files and `report.txt` go (created if missing)
`--prefix`	No	The motion name prefix, e.g. `FitPeachBody`; read from `main.pac` automatically
`file.anim`	At least one	An animation to convert; its motion name comes from the file name
`file.anim=MotionName`	Optional form	Puts that animation into a different motion slot

Skirt handling (`--skirt`)

The Peach/Daisy skirt is built differently in Ultimate, so the script picks a skirt source per animation. The default, `auto`, chooses for you and logs its choice at the top of each motion in `report.txt`.

Mode	What the skirt does	When to use it
`auto` (default)	`stock` when the converted legs match the stock move (same length, legs within 8°), `stock + push-out` when the length matches but the legs differ, `otherwise anchored + push-out`	Almost always
`stock`	Keeps Smash 4's own skirt motion for that move	Moves Ultimate reused from Smash 4, like dash and run
`anchored`	Ultimate's sway around Smash 4's natural skirt shape	New moves, like Daisy's idles
`ult`	Ultimate's skirt rotations as they are	Only for comparison; this is what caused leg clipping and a raised back hem

Push-out. With `anchored` skirts, and stock skirts under legs that move differently, the script also checks every frame for a knee, shin or shoe poking through the dress. Where one does, it swings the nearest skirt panels outward just enough to cover the leg, easing in and out over a few frames. `report.txt` lists the frame counts before and after for each motion. Add `--no-pushout` to turn it off.

Motion names

The file name sets the slot: `a00wait1.anim` becomes `A00Wait1`, `a02run.anim` becomes `A02Run`. Matching ignores upper and lower case.

When the Ultimate and Smash 4 names differ, or you want an animation in a second slot, add `=` and the Smash 4 name. The same file can be listed more than once:

```
... a00wait2.anim a00wait2.anim=A00Wait4
```

That puts Ultimate's Wait2 into both Smash 4's Wait2 and Wait4.

What it prints

One line per motion per .pac, also saved to `report.txt`:

```
A02Run: main.pac FitPeachBodyMainA02Run.omo  33 frames (stock 33), 34 bones,
8 kept from stock
```

- **frames (stock N):** the new length next to the original Smash 4 length.
- **kept from stock:** bones Smash 4 animates that Ultimate has no match for; they keep their original motion, stretched to the new length.
- **has no ... skipped:** that .pac has no motion by that name. Check the spelling or use `=MotionName`.

Step 4: Install and test

Copy the new .pac files into the workspace, rebuild the patch, and check the motions in game.

1. Back up the fighter's current files in `workspace/content/patch/data/fighter/<name>/motion/body/`.
2. Copy every `.pac` from your `--out` folder into that folder, replacing the old ones.
3. Rebuild the patch in Sm4shExplorer as you normally do.
4. Start a match in Cemu and watch each converted motion: idle for the waits, hold a direction for dash and run.

The motion .pac files are shared by every costume of the fighter, so the new animations show on all slots.

To undo: copy the original .pac files back from your backup, or from `extract/data/fighter/<name>/motion/body/` for stock, and rebuild.

How the conversion works

For each animation the script runs the same six steps, then writes one .omo per .pac.

1. **Read the keys.** It takes the `.anim` keyframes 1 to N. StudioSB sets the end time to N+1, so N matches the frame count Smash 4 uses for the same move.
2. **Rename bones.** Ultimate names become Smash 4 names using the reverse of the original tool's 288-pair table, plus `HaveL/R`, `Throw`, `Trans`, `Rot`, `Hip` and the Peach/Daisy swing bones.

3. **Fold the extra bones.** Ultimate adds `ClavicleC` above the shoulders and `LegC` above the legs. Their motion is merged back into `LShoulderN` / `RShoulderN` and `LLegJ` / `RLegJ`, so the limbs end up in the same place.
4. **Convert rotations.** Euler angles become quaternions using the order R = Rz · Ry · Rx, the order both games share.
5. **Handle rig differences.** Bones whose Ultimate rig was rebuilt (the Peach/Daisy skirt) keep Smash 4's bone lengths and take only Ultimate's rotations. Face bones are skipped, since Smash 4 body animations don't animate them.
6. **Split by .pac.** Each .pac gets exactly the bones its stock .omo animated: body bones to main.pac, fingers to lhand/rhand.pac, swing bones to the swing pacs. A bone Ultimate lacks keeps its stock motion, stretched to the new length.

The .omo files use the same 16-bit keyframe layout as Smash Forge, which already ran in game for the stage work. The .pac writer rebuilds stock .pac files byte-for-byte, so nothing else in them shifts.

Limits and troubleshooting

Idles and movement port cleanly. Attacks, skirts and other rigs need care.

Problem	Cause	What to do
Report says `has no ... skipped`	That .pac has no motion with that name	Check the spelling, or name the slot with `file.anim=MotionName`
Nothing changed in game	The patch wasn't rebuilt, or the pacs went to the wrong folder	Copy to `motion/body/` and rebuild in Sm4shExplorer
Hitboxes don't line up with an attack	Timing now follows Ultimate, but Smash 4's `game.bin` still uses the old frames	Stick to idles and movement, or retime that move's script
Animation is longer or shorter than before	Length follows Ultimate	Fine for loops and idles; the game plays the whole thing
Skirt or hair moves differently	Ultimate rebuilt those swing bones	Expected; the top of the skirt follows closely, the lower parts are approximate

Problem	Cause	What to do
Arms or legs twisted badly	The two skeletons don't match for this fighter	Check bone names; see the next section
Many bones `kept from stock`	Ultimate bone names didn't map	Add the missing names to the table (next section)
Script fails to read a `.anim`	Export missing keyframes or bones	Re-export with all bones and every frame

- **Every costume changes.** Motion pacs are shared by all slots of the fighter.
- **Face expressions don't carry over.** Smash 4 handles faces with separate expression files, not body animations.

Using it for other fighters

Body and finger bones already map for every fighter the original S4-Anims-to-Ult covers. Swing bones (hair, capes, tails, skirts) are named per fighter, so each new fighter needs its own lines added.

Find the names.

- **Ultimate:** open a `.anim` in a text editor. Each bone appears in lines like `anim rotate.rotateX rotateX S_Hair1 ...`.
- **Smash 4:** open the fighter's `model.vbn` in Smash Forge and read the bone list. Swing bones look like `SWG_Bhair0__swing`.

Add the lines. Near the top of `ult2s4.py`, under the existing hair line, add one line per bone:

```
U2S['S_Cape1'] = 'SWG_Cape0__swing'
U2S['S_Cape2'] = 'SWG_Cape1__swing'
```

Those cape names are made-up examples; use the real ones you found.

- **Rig rebuilt in Ultimate?** If the bone chain has different lengths in Ultimate, also add it to `ROT_ONLY` so Smash 4's lengths are kept: `ROT_ONLY.add('S_Cape1')`.
- **Extra in-between bone?** If Ultimate inserts a bone that Smash 4 doesn't have, add it to `FOLD` as `child: inserted parent`, the way `ClavicleL: ClavicleC` is set up.

Run the script and read `report.txt`. A drop in `kept from stock` means the new names mapped.

Worked example: Daisy onto Peach

Seven Peach motions were replaced with five of Daisy's Ultimate animations in one run. Daisy has only three idles, so her Wait2 and Wait3 also fill Peach's Wait4 and Wait5.

```
python ult2s4.py --vbn peach_check\c09\model.vbn --pacs pacs --out out_peach
a00wait1.anim a00wait2.anim a00wait3.anim a00wait2.anim=A00Wait4
a00wait3.anim=A00Wait5 a02dash.anim a02run.anim
```

Peach motion	Daisy file	New frames	Stock frames
A00Wait1	a00wait1.anim	236	101
A00Wait2	a00wait2.anim	94	240
A00Wait3	a00wait3.anim	130	270
A00Wait4	a00wait2.anim	94	190
A00Wait5	a00wait3.anim	130	125
A02Dash	a02dash.anim	35	35
A02Run	a02run.anim	33	33

- **Accuracy:** Daisy's dash matched Peach's stock dash within 0–5° on the body bones, since Ultimate reused that animation.
- **Loops:** the idles start and end on the same pose, so they loop without a pop.
- **Files:** the script, these pacs and the report are saved in `Claude Stage Projects\Ult2S4\`.